

Lecture 24: Testing of Hypotheses - the NP Lemma

One more example before introducing the NP Lemma:

Example 2

The standardized lifetime of an equipment is normally distributed with mean θ & std. dev. 5. For testing $H_0: \theta \leq 30$ ag. $H_A: \theta > 30$, a random sample of size n is chosen. Determine the sample size n and the cut-off value c of the test function.

$$\phi(\bar{x}) = \begin{cases} 1 & \text{if } \bar{x} \geq c \\ 0 & \text{if } \bar{x} < c \end{cases}$$

has power β_ϕ equal to 0.1 and 0.9 at $\theta = 30$ and 35, respectively.
(Given $\Phi(1.282) = 0.9$).

As in Ex 1, we consider the power β_ϕ first, but in terms of the test ϕ .

Thus, power $\beta_\phi(\theta) = E_\theta \phi(\bar{X}) = P_\theta(\bar{X} \geq c) = 1 - P(\bar{X} < c)$

$$= 1 - \Phi\left(\frac{\sqrt{n}(c - \theta)}{5}\right)$$

$$\left. \begin{array}{l} \text{Given, at } 30, \beta_\phi(30) = 0.1 \\ \quad \quad \quad 35, \beta_\phi(35) = 0.9 \end{array} \right\} \Rightarrow \left. \begin{array}{l} \Phi\left(\frac{\sqrt{n}(c - 30)}{5}\right) = 0.9 = \Phi(1.282) \\ \Phi\left(\frac{\sqrt{n}(c - 35)}{5}\right) = 0.1 = \Phi(-1.282) \end{array} \right\}$$

$$\Rightarrow \frac{\sqrt{n}(c - 30)}{5} = 1.282 \text{ and } \frac{\sqrt{n}(c - 35)}{5} = -1.282$$

$$\text{Giving } c = 32.5, n \approx 7$$

(Try to comprehend the implications of the two examples, it's a bit, the theory learnt so far.)

The Neyman-Pearson Lemma (For continuous distns) - Statement only, no proof.
(Case of discrete distributions follows clearly from here.)

Let $X \sim f_0, \theta \in \Theta$.

$g_0, g_1, g_2, \dots$ are integrable functions defined over the sample space $\mathcal{X}$.

Let $W \subset \mathcal{X}$, be a region $\ni \int_W g_i dx = c_i$ (fixed), $i = 1, 2, \dots$ (*)

Further $\exists$ constants $k_1, k_2, \dots \ni g_0 \geq \sum_{i=1}^{\infty} k_i g_i$ within W_0 (where W_0 satisfies (*))
 $<$ outside W_0 .

Then $\int_{W_0} g_0 dx \geq \int_W g_0 dx$ for W satisfying (*).
 Next: Application of NP Lemma in constructing MP & UMP sized tests.